F18CE2 - MV Calculus and Real Analysis B: Tutorial 5

1 State:

(a) the limit definition of f is continuous at a;

(b) the $\epsilon - \delta$ definition of f is continuous at a.

2 Prove that f is discontinuous at 0, where $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 3x^2 + 1, & \text{if } x \geqslant 0, \\ x + 2, & \text{if } x < 0. \end{cases}$$

3 Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2 + 3, & \text{if } x \geqslant 1, \\ x^3 + 2, & \text{if } x < 1. \end{cases}$$

Is f continuous at $x = 1$?

4 Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} ax^2 + bx, & \text{if } -1 \leqslant x \leqslant 1, \\ cx + d, & \text{otherwise,} \end{cases}$$

where a, b, c, d are constants. Find conditions on a, b, c, d so that f is continuous on $\mathbb{R}$.

5 Prove that f is continuous at 0, where $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

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6 Prove that f is discontinuous at 0, where $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0, \\ 1, & \text{if } x = 0. \end{cases}$$

7 Prove that f is discontinuous at 0, where $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} (x+1) \cos \frac{1}{x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

8 Let $f : \mathbb{R} \to \mathbb{R}$ be continuous at 0 with $f(0) = 0$. Let $g : \mathbb{R} \to \mathbb{R}$ satisfy $|g(x)| \leqslant |f(x)|$ for all x. Prove that g is continuous at 0.

9 Let f be continuous at a with $f(a) > 0$. Prove that there exists $\delta > 0$ such that

$$|x - a| < \delta \implies f(x) > \frac{1}{2} f(a) > 0.$$

10 Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x + y) = f(x) + f(y)$ for all x and y in $\mathbb{R}$.

Show that $f(0) = 0$.

If f is continuous at 0, prove that f is continuous at all $a \in \mathbb{R}$.

11 Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)\,f(y)$ for all x and y in $\mathbb{R}$ and $f(0) \neq 0$.

Show that $f(0) = 1$.

If f is continuous at 0, prove that f is continuous at all $a \in \mathbb{R}$.

12 State the Intermediate Value Theorem. Prove that the equation $x^7 - 4x^4 + x + 1 = 0$ has a solution c with $0 < c < 1$.

13 Prove that the equation $xe^x + \sin^2 x = 1$ has a solution in $\mathbb{R}$.

14 Suppose $f : [0,1] \to \mathbb{R}$ is continuous on $[0,1]$ and $f(0) > 0,\ f(1) < 1$. Show that the equation $f(x) = x$ has a solution in $[0,1]$.

15 Suppose $f : [0,1] \to \mathbb{R}$ is continuous on $[0,1]$ and $f(0) = f(1)$. Prove that there exists $y \in [0, 1/2]$ with $f(y) = f(y + 1/2)$.

16 Give an example of a function $f : [0,1) \to \mathbb{R}$ which is continuous but not bounded on $[0,1)$.